

A Categorical Perspective on Kripke Models and Dynamic Epistemic Logic

Haitian Wang

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Outline

Dynamic Epistemic Logic

Categories of Kripke Models

Update Actions as Functors

Ongoing Work

Overview

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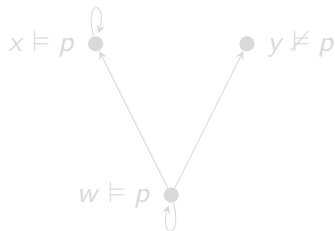
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Epistemic Logic

In modal logic, we have the $\Box$ operator to represent necessity:

$$M, x \models \Box \varphi \text{ iff } M, y \models \varphi \text{ for all } y \text{ s.t. } xRy$$

In epistemic logic, we interpret $\Box$ as an epistemic/doxastic operator:



► $w \not\models \Box p$

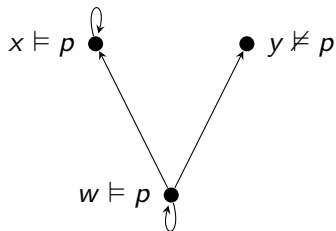
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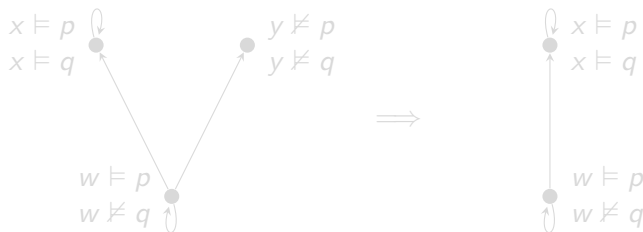
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Dynamic Operators as Updates

We could study the dynamic aspects such as belief/knowledge revisions by adding the dynamic operators.

$M, x \models [!\psi]\varphi$ iff $M', x \models \varphi$ where M' is the submodel derived by restricting to $\llbracket \psi \rrbracket_w$.



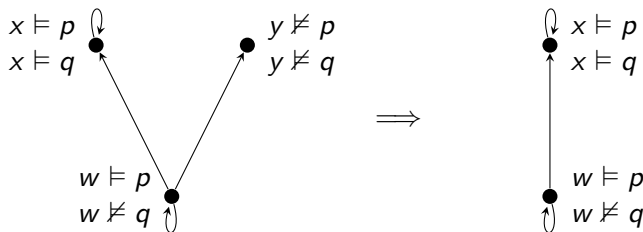
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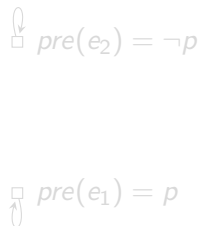
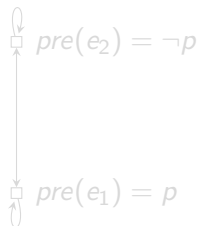
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Action Models and Product Updates

More general updates are modeled by action models:

An action model is a structure (A, R_A, pre_A) , where (A, R_A) is a Kripke frame and pre_A assigns to each action $a \in A$ a formula $pre_A(a)$ that is interpreted as the precondition of the action a .

Two examples:

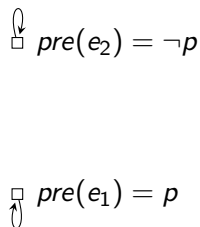
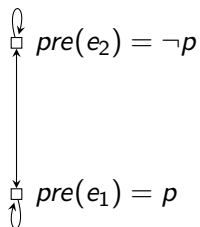


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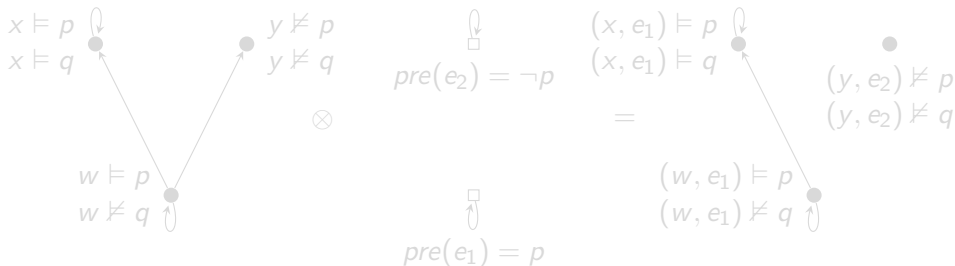


Action Models and Product Updates

Given a Kripke model $M_X = (X, R_X, V_X)$ and an action model (A, R_A, pre_A) , the product update model is $M_{X \otimes A} = (X \otimes A, R_{X \otimes A}, V_{X \otimes A})$ where

- ▶ Worlds: $X \otimes A = \{(x, a) \mid x \models pre_A(a)\}$,
- ▶ Relation: $(x, a)R_{X \otimes A}(x', a')$ iff xR_Xx' and aR_Aa'
- ▶ Valuation: $(x, a) \in V_{X \otimes A}(p)$ iff $x \in V(p)$.

$$M_X, w \models [E, e]\varphi \text{ iff } M_{X \otimes A}, (w, e) \models \varphi.$$



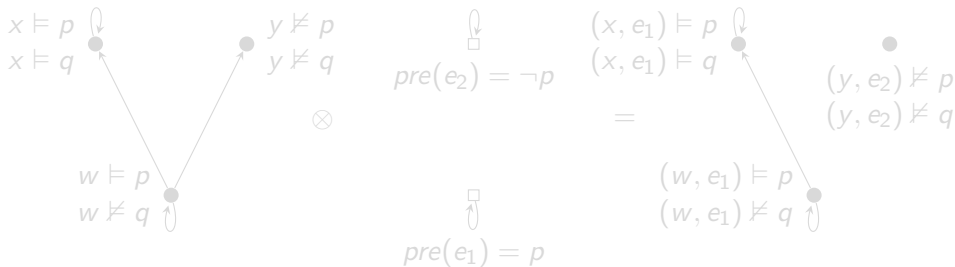
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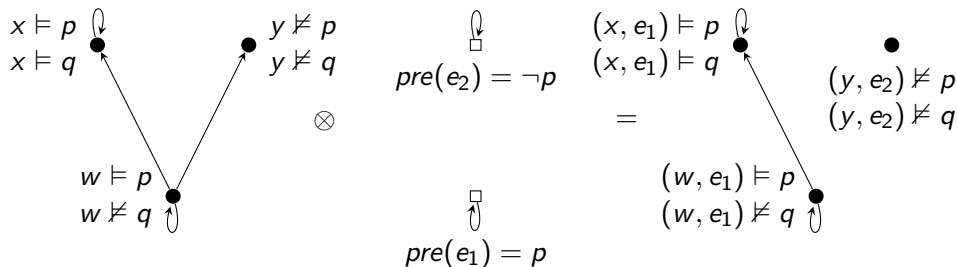
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Homomorphisms as Morphisms

Definition (Categories of Kripke frames/models)

A Kripke frame (X, R) is a set together with a binary relation. A homomorphism $f : (X, R_X) \rightarrow (Y, R_Y)$ is a function such that for all $w, v \in X$, $wR_X v$ implies $f(w)R_Y f(v)$. The category of Kripke frames and their homomorphisms is denoted by $\mathbb{K}_{\mathbb{F}}$.

This definition can be extended to model-level also: Given a fixed set P of propositional variables, a Kripke model (X, R_X, V_X) is a Kripke frame (X, R_X) together with a valuation $V_X : P \rightarrow \mathcal{P}(X)$. A homomorphism $f : (X, R_X, V_X) \rightarrow (Y, R_Y, V_Y)$ at model-level is a homomorphism between frames that preserves the valuation: $w \in V_X(p)$ implies $f(w) \in V_Y(p)$. The category of Kripke models is denoted by $\mathbb{K}$.

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Different Morphisms

By requiring more properties of homomorphisms, we have different choices of morphisms:

Morphisms	Formulas preserved
homomorphisms	positive formulas (built using p , $\Diamond$, $\wedge$, $\vee$)
prop-reflecting homs	existential formulas (built using p , $\neg p$, $\Diamond$, $\wedge$, $\vee$)
p-morphisms	all modal formulas

Table: Morphism types and the classes of formulas they preserve

An advantage of using homomorphisms as morphisms:

$\mathbb{K}$ and $\mathbb{K}_{\mathbb{F}}$ are topological over Set . This is useful when giving a categorical semantics to DEL (Kishida 17, Ye 22).

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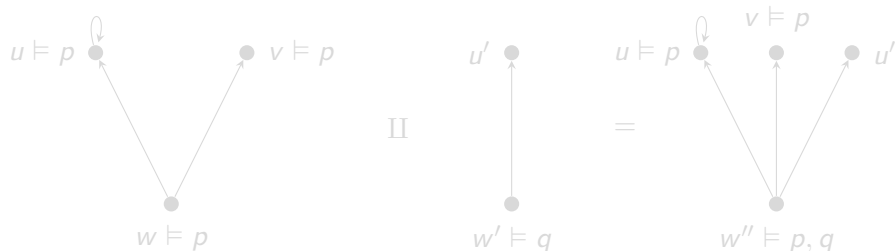
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Different Objects

A pointed Kripke model (X, R_X, V_X, w) is a Kripke model with a designated world. An epistemic interpretation would see w as the actual world and the accessibility relation as indistinguishability relation. Modal formulas true at w represent the epistemic status of the agent. A morphism $f : (X, R_X, V_X, w) \rightarrow (Y, R_Y, V_Y, w')$ is a homomorphism that preserves the actual world, i.e. $f(w) = w'$.

We give an example of coproduct in this category of pointed Kripke models:

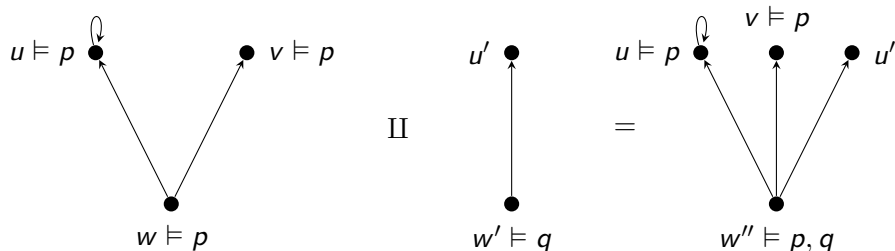


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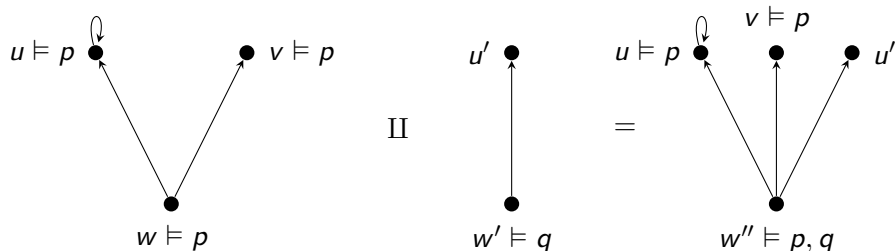


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Action models as operations

Given any action model A , an operation $- \otimes A$ is naturally induced:

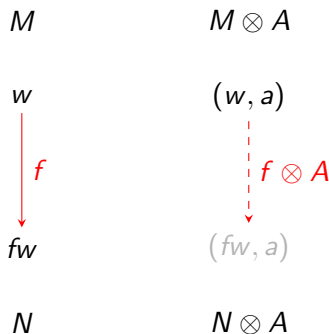
Definition (Action operations)

For an action model A , define an operation $- \otimes A : \mathbb{K} \rightarrow \mathbb{K}$ by:

- ▶ on objects: $M \mapsto M \otimes A$,
- ▶ on morphisms: for a homomorphism $f : M \rightarrow N$, let $f \otimes A : M \otimes A \rightarrow N \otimes A$ send (w, a) to $(f(w), a)$.

How the natural idea fails

We expect action operations to work as functors but they are not well-defined in general:



The main obstruction is that homomorphisms need not preserve modal formulas in general (Table 2): it may happen that $w \models \text{pre}_A(a)$ but $f(w) \not\models \text{pre}_A(a)$. Thus, the assignment $(w, a) \mapsto (f(w), a)$ may send a valid pair in $M \otimes A$ to a non-valid pair in $N \otimes A$, so $f \otimes A$ may fail to be a morphism¹.

¹A modification of product update to keep all the pairs (w, a) after update and to only remove the relations will not solve the problem as the relation condition will still be violated.

An ad hoc Fix

We can restore functoriality by restricting morphisms or preconditions:

Morphisms	Formulas preserved
homomorphisms	positive formulas (built using $p, \Diamond, \wedge, \vee$)
prop-reflecting homs	existential formulas (built using $p, \neg p, \Diamond, \wedge, \vee$)
p-morphisms	all modal formulas

Table: Morphism types and the classes of formulas they preserve

One can check that restricting preconditions indeed restores functoriality of $- \otimes A$ for the given choice of morphisms.

We also obtain interesting structures. For example, $(\mathbb{A}_{Pos}, ;, skip)$ is a monoidal category and product update $M \triangleleft A := M \otimes A$ equips $\mathbb{K}$ with the structure of a (right) $\mathbb{A}_{Pos}$ -actegory.

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A non-ad hoc fix using Kleisli Category

A different approach is to work in a Kleisli-style category encoding partial maps. The usual encoding of partial maps into N as total maps into $N + 1$ fails in $\mathbb{K}$, because the coproduct $N + 1$ does not guarantee the relations being preserved. We therefore use a *dummy-state* construction.

Definition (Dummy-state functor and its Kleisli category)

Define an endofunctor $T : \mathbb{K} \rightarrow \mathbb{K}$ as follows. For a Kripke model $M = (W, R, V)$, let $T(M) := (W + 1, R^T, V^T)$, where

- ▶ $W + 1 = W \sqcup \{\perp\}$
- ▶ $R^T := R \cup \{(\perp, x) \mid x \in W + 1\} \cup \{(x, \perp) \mid x \in W + 1\}$
- ▶ $V^T(p) := V(p) \cup \{\perp\}$

For a morphism $f : (W, R, V) \rightarrow (W', R', V')$ in $\mathbb{K}$, define

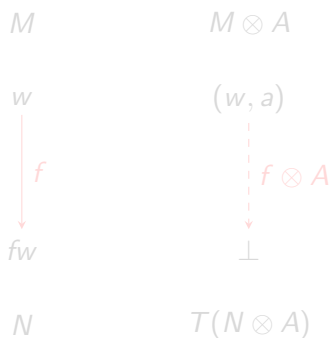
$T(f) : W + 1 \rightarrow W' + 1$ by

- ▶ $T(f)(w) := f(w)$ for $w \in W$
- ▶ $T(f)(\perp) := \perp$

The induced Kleisli category $\mathbb{K}_T$ has the same objects as $\mathbb{K}$, and for morphisms $\text{Hom}_{\mathbb{K}_T}(M, N) := \text{Hom}_{\mathbb{K}}(M, TN)$.

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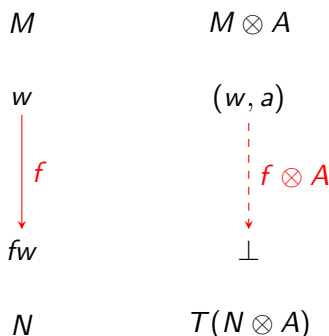
The product update is functorial when regarded as a partial operation. The dummy world $\perp$ represents failure of an action to apply at a world. In this setting, a product update $- \otimes A$ that is not an endofunctor on $\mathbb{K}$ becomes a functor from $\mathbb{K}$ to $\mathbb{K}_T$.



The extra requirements about valuations and relations make sure that $f \otimes A$ preserves the monotonicity requirement of a homomorphism in $\text{Hom}_{\mathbb{K}_T}(M \otimes A, N \otimes A) := \text{Hom}_{\mathbb{K}}(M \otimes A, T(N \otimes A))$.

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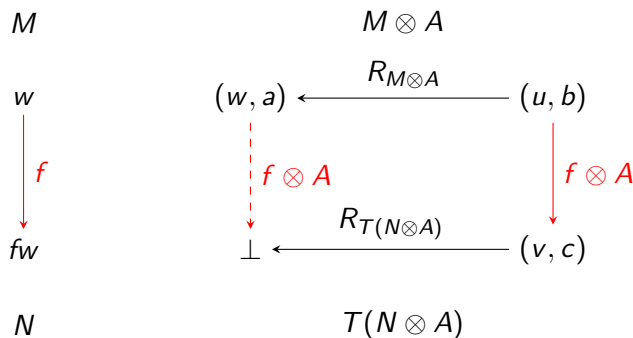
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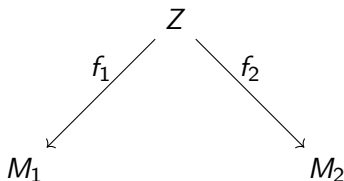
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Generalize concepts to multi-pointed Kripke models

Bisimulation

For single-pointed Kripke models:

M_1 and M_2 are bisimilar $\Leftrightarrow$ there exists a span of open maps (p-morphisms) f_1 and f_2 (Joyal, Nielsen and Winskel, 1996):



This can be generalized to multi-pointed Kripke models. With the same path category and notion of open map, adapting the morphisms to homomorphisms in the multi-pointed case yields the corresponding notion of bisimulation ².

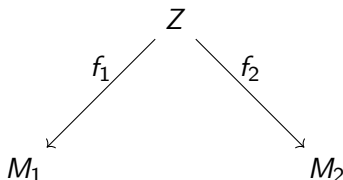
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From tree unravellings to Forest Unravellings

For single-pointed Kripke models, a tree unravelling is obtained by 'unravelling' the model with respect to the single root.

For a multi-pointed Kripke model, we obtain the forest unravelling of it by doing the usual tree unravelling with respect to each actual world take the disjoint union of the trees to form a forest.

Inspired by the homotopical view of modal logic in (Reggio, 2023), we expect this construction to be relevant when we explore the homotopic aspects of multi-pointed Kripke models and action models in the future.

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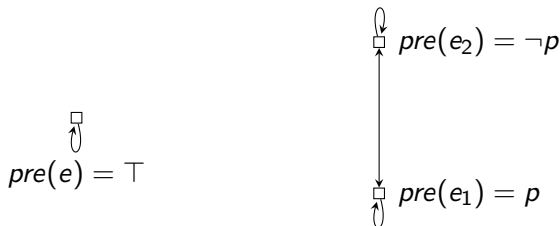
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Action Emulation

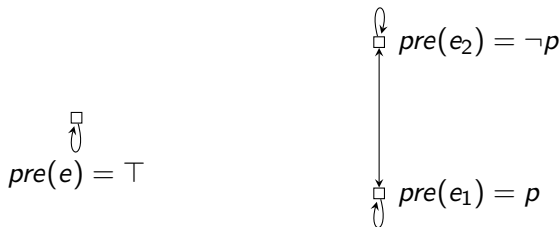
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The induced equivalence relation on action models is called *action emulation* and is characterized in (van Eijck, Ruan and Sadzik, 2012). That construction is formulated in a single-pointed setting and is sensitive to the choice of designated state. Our categorical and multi-pointed viewpoint suggests a more structural account of action emulation, and we plan to investigate it from a homotopical perspective.

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- [1] Kishida, K. (2017). Categories for dynamic epistemic logic. TARK2017.
- [2] Ye, L. (2022). Unification of Modal Logic via Topological Categories. ACT2022.
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